Exercise Session – Chapter 4: SDN, NFV, IoT

Exercise 1

Explain in your own words the three main differences between DV and LS routing protocols:

- Information exchanged
- Visibility of the network
- Convergence behavior

Exercise 2

List and briefly describe any five OSPF neighbor states.
Explain why OSPF needs multiple states instead of directly going to Full.

Exercise 3

Give three differences between **internal routing protocols (like OSPF)** and **BGP** regarding:

- Purpose
- Routing information
- Convergence

Exercise 4

Two OSPF routers are stuck in **Init state**.
You check the hello packets and find:

Parameter	**R1**	**R2**
Area ID	0	0
Hello interval	10s	10s
Dead interval	40s	40s
Network type	Broadcast	Broadcast
Authentication	None	None
MTU	1500	1400

Identify the reason why adjacency is not forming and explain why MTU matters.

Exercise 5

A router receives **three BGP routes** to reach 10.5.0.0/16:

Attribute	**Path A**	**Path B**	**Path C**
LOCAL_PREF	300	200	300

AS_PATH	65001 65010	65002	65001 65010
ORIGIN	i	i	e
MED	—	—	—
Type	eBGP	iBGP	eBGP
IGP cost to next-hop	5	3	12
Router ID	3.3.3.3	1.1.1.1	2.2.2.2

Exercise 6

Link costs (OSPF cost = 100 / bandwidth in Mbps):

- R1 → R2: 1 Gbps
- R2 → R3: 100 Mbps
- R1 → R3: 10 Mbps

Questions:

1. Calculate the OSPF cost for each link.
2. Which path will OSPF choose from R1 to R3?
3. Explain why OSPF does not simply choose the path with fewer hops.

Exercise 7

Switches have the following priorities and MACs:

Switch	Priority	MAC Address
SW1	32768	AA-AA-AA-AA-AA-01
SW2	24576	AA-AA-AA-AA-AA-02
SW3	32768	AA-AA-AA-AA-AA-00

1. Which switch becomes the root bridge?
2. Why does SW3 beat SW1 even with the same priority?
3. What happens if you manually set SW1 priority to 4096?

Exercise 8

- Why does OSPF require matching hello/dead timers?
- Why does BGP not use multicast/broadcast to discover neighbors?
- Why do switches use MAC tables but routers use routing tables?
- Why is the backbone area required in OSPF?
- Why do we avoid running BGP on every internal router?

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SOLUTIONS

Exercise 1

Three main differences

1. **Information exchanged**
   - **DV:** Routers periodically send their entire routing table to neighbors (destination + metric + next-hop).
   - **LS:** Routers send *link-state advertisements (LSAs)* describing their local links; everyone builds a full topology (LSDB).
2. **Network visibility**
   - **DV:** Each router only knows its neighbors' view (no global map).
   - **LS:** Each router has a synchronized map of the area/network (full topology).
3. **Convergence behavior**
   - **DV:** Slower convergence and susceptible to loops (count-to-infinity).
   - **LS:** Faster convergence (event-driven LSAs + SPF) and loop-free (SPF guarantees shortest-path tree).

Exercise 2

List & describe five states (brief):

1. **Down** — No Hello received yet from that neighbor.
2. **Init** — Hello received from neighbor but neighbor's Hello did not list this router in its neighbor list. (One-way)
3. **2-Way** — Bidirectional communication confirmed (each sees the other in Hello neighbor lists). DR/BDR election may occur here.
4. **ExStart** — Beginning of DB synchronization: master/slave negotiation and sequence number setup.
5. **Exchange** — Routers send Database Description (DBD) packets summarizing their LSDBs.
6. **Loading** — Router requests missing LSAs (LSR) and receives LSUs.

7. **Full** — LSDBs synchronized and adjacency complete.

Why multiple states?
They enforce ordered, reliable synchronization of topology data (avoid inconsistencies). Each state handles a specific phase (discovery → verification → DB sync → full readiness), improving robustness and error handling.

Exercise 3

Three differences:

1. **Purpose**
   - **OSPF:** IGP used *inside* an AS for shortest-path routing (fast convergence).
   - **BGP:** EGP used *between* ASes for policy-based path selection.
2. **Routing information**
   - **OSPF:** Exchanges link-state info and builds full topology (LSDB).
   - **BGP:** Exchanges path information and attributes (AS_PATH, MED, COMMUNITY); focuses on policies.
3. **Convergence & control**
   - **OSPF:** Fast SPF-based convergence; deterministic shortest paths.
   - **BGP:** Slower convergence; chooses paths based on policy attributes rather than pure shortest path.

Exercise 4

Reason adjacency not forming: MTU mismatch (R1 MTU 1500, R2 MTU 1400).

Why MTU matters:

- During DB Exchange (ExStart/Exchange) routers exchange DBD packets and then request LSAs. If a router sends an LSA larger than the peer's MTU, the peer may fail to accept or properly parse packet fragments, causing synchronization to stall. OSPF peers check interface MTU during ExStart; mismatch prevents full adjacency for safety (to avoid truncated LSAs / inconsistent LSDBs).

Fix: Configure both interfaces to same MTU or enable consistent fragmentation/adjustment so DBD/LSU exchange can proceed.

Exercise 5

BGP Path Selection (table)

Given the three paths, decide:

Attributes recap:

- LOCAL_PREF: A=300, B=200, C=300 → A & C tied (highest)
- AS_PATH: A = 65001 65010 ; C = 65001 65010 → equal
- ORIGIN: A = i (IGP), C = e (EGP) → i better than e
  So **Path A is selected** at step 3 (ORIGIN).

Step-by-step decision:

1. LOCAL_PREF eliminates B (200 vs 300).
2. Compare A vs C: AS_PATH equal → no decision.
3. ORIGIN: A (i) preferred over C (e) → A wins. (No need to check MED, eBGP/iBGP preference, IGP cost, etc.)

Exercise 6

OSPF Metric Calculation

Given cost formula used in slides: **Cost = 100 / bandwidth(Mbps)** (reference 100 Mbps).

1. Link costs:
   - R1 → R2: 1 Gbps = 1000 Mbps → Cost = 100 / 1000 = 0.1 → typically rounded to **1** (most implementations clamp to at least 1).
   - R2 → R3: 100 Mbps → Cost = 100 / 100 = **1**.
   - R1 → R3: 10 Mbps → Cost = 100 / 10 = **10**.
     (Note: Cisco uses reference-bandwidth/if-bandwidth; default RB=100 Mbps → same results; many platforms round to integer ≥1.)
2. Path costs:
   - R1 → R2 → R3 cost = 1 + 1 = **2**.
   - R1 → R3 direct cost = **10**.
     → OSPF chooses **R1→R2→R3** (cost 2 is smaller).
3. **Why not fewer hops?**
   - OSPF optimizes *cost* not hop-count. A path with fewer hops can have lower bandwidth and thus higher cost. OSPF metric encodes link speed so a multi-hop high-speed path can be preferred.

Exercise 7

Given switches:

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Switch	Priority	MAC
SW1	32768	..01
SW2	24576	..02
SW3	32768	..00

1. **Which becomes root?**
   - Lowest bridge ID wins. Bridge ID = Priority + MAC. SW2 has lowest priority (24576 < 32768), so **SW2 becomes root**.
2. **Why does SW3 beat SW1 when same priority?**
   - When priorities equal (32768), lowest MAC addresses decide. SW3 MAC ..00 < SW1 MAC ..01 → SW3 would win between those two.
3. **If SW1 priority set to 4096**
   - Lower priority values have higher precedence. Setting SW1 to 4096 makes it the lowest priority (4096 < 24576 and < 32768?) — actually 4096 < 24576, so **SW1 becomes the root**. (Note: lower numeric priority = preferred root.)

Exercise 8

Short answers:

1. **Why does OSPF require matching hello/dead timers?**
   - To ensure both routers speak the same timing/expectation; mismatched timers can cause false neighbor timeouts or unexpected drops, leading to instability. Matching guarantees agreed heartbeat intervals and failure detection timing.
2. **Why does BGP not use multicast/broadcast to discover neighbors?**
   - BGP peers are explicitly configured and use TCP (reliable). Discovery via multicast would be inappropriate for interdomain policy-based routing and would create security/scale issues. BGP is point-to-point by design.
3. **Why do switches use MAC tables but routers use routing tables?**
   - Switches operate at Layer 2 and forward frames based on MAC addresses learned on ports (local LAN switching). Routers operate at Layer 3 and forward packets based on network prefixes and next hops (routing tables).
4. **Why is the backbone area required in OSPF?**
   - Area 0 provides the transit for inter-area routing; it organizes routing hierarchy, prevents loops, and limits LSA scope. Without Area 0, inter-area summaries and routing would be inconsistent.

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5. **Why do we avoid running BGP on every internal router?**
    - BGP sessions and policies are heavyweight and complex; running BGP everywhere creates management and scaling problems. Typically only border routers and selected internal route reflectors run BGP to control complexity and ensure policy enforcement.